Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
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Project Title:	Space Flight Immunodeficiency		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Immunology, Infection and Hematology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Conner, Margaret (Baylor College of Medicine) Butel, Janet (Baylor College of Medicine) Lednicky, John (Baylor College of Medicine) Ling, Paul (Baylor College of Medicine) Pierson, Duane (NASA JSC) Lugg, Desmond (Australian Antarctic Division) Ochs, Hans (University of Washington) Reuben, James (The University of Texas M.D. Anderson Cancer Center) Rosenblatt, Howard (Baylor College of Medicine) Smith, E. (Baylor College of Medicine) Nelson, Gregory (Loma Linda University) Lee, Bang-Ning (The University of Texas M.D. Anderson Cancer Center) Gridley, Daila (Loma Linda University)		
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Task Description:	On January 14, 2004, the President of the United States announced a new vision for the National Aeronautics and Space Administration (NASA). NASA included the directives to implement a human program to explore the solar system and beyond and extend human presence across the solar system, starting with human return to the Moon by the year 2020, in preparation for human exploration of Mars and other destinations. Long-term interplanetary space flight, such as the journey to the planet Mars envisioned within the next few decades, poses substantial health risks for human. Many space flight conditions as tested in human and animal space travelers, and in earthbound human and animal models of space flight give ample evidence of compromised immunity and the inability to resist microbial infections. Perhaps, the most serious physical threat to health in space is radiation that is composed of photons (X-rays), electrons (g-rays), protons, neutrons, and heavy metal ions. Investigations of mice given either 3 Gy of radiation (protons or g-rays) demonstrated that by days 4 to 10 after irradiation, statistically significant decreases in CD19+ B cells, CD3+ T cells, CD4+ T cells, and CD8+ T cells, and IL-2 secretion by activated spleen cells. Irradiated animals showed a delayed and lowered anti-sRBC antibody response. Natural killer cells were relatively radioresistant. B cells recovered by day 15 and CD3+ and CD4+ T cells by day 29, but CD8+ T cells remained impaired. As has been learned in transplant medicine, reactivation of latent viruses from within the host pose a formidable threat when host immunity is compromised. Latent viruses shedding has been shown to increase in humans in space shuttle flights (Epstein-Barr virus; cytomegalovirus) and in the Antarctic winter-over model of space flight (polyomavirus, herpesvirus). These viruses are known to be associated with human and animal cancers. 1) Original Aims of Project The hypothesis being addressed in our studies reported here is that space flight radiation will suppress the human immune system leading to reactivation of latent viruses, increased viral infections and disease, and the development of cancers. Since we can never purposefully test this hypothesis in humans, we use a murine animal model of g-ray and latent polyomavirus (PyV) infection to determine harmful effects upon the immune system with reactivation of latent virus infection. 2) Key Findings of Project Groups of BALB/c female mice were given whole body irradiation (3 Gy 137Cs) or sham irradiation on day 0 and 49, and murine polyomavirus (PyV) or saline control on day 1: A, 3 Gy + PyV; B, no Gy + PyV; C, 3 Gy + no PyV; and D, no Gy + no PyV. Mice were tested for PyV replication by quantitative PCR and spleen weights and cell counts, and proliferation and gamma interferon (IFN-g) production were measured at various intervals up to 69 days. Group A showed elevated PyV replication on days 10 and 20, as compared to Group B and both Groups A and B cleared PyV by day 49 in A and 20 in B. Only Group A again showed PyV replication when given a second dose of radiation on day 49. Spleen weights and cell counts of Group A were significantly lower than other groups. Irradiation suppressed T-cell proliferation in Groups A, B and C except in Group B when PyV was cleared. PyV infection enhanced IFN-g in all Groups: B > A > C. (3) Impact of Findings on Hypotheses, Objectives, and Specific Aims of Original Project This model of space flight suggests that the combined effects of radiation and latent virus infection will severely affect T-lymphocyte mediated immunity that may lead to chronic viral infection and malignancy. Thus, these findings partially validate the hypotheses, complete the objectives, and reply to the specific aims of the original project. (4) Proposed Research Plan for Coming Year Our plan for the coming year is to publish at least one, possibly two, manuscripts of this research in a high-quality, peer-reviewed science journal. Also, because of the extraordinary expense of animal work, we plan to turn to an in vitro B cell model to investigate the effects of radiation and countermeasures that could reverse immunosuppression. We will be able to perform many more experiments at a fraction of the cost of the animal experiments. When we have more information on mechanisms of apoptosis and its reversal, we plan to return to the animal model for selected experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Humans undergoing multi-modal immunosuppression for organ or bone marrow stem cell transplantation occasionally (i.e., 2% occurrence) develop polyclonal activation of B cells due to the latent Epstein-Barr virus (EBV). Repetitive activation of B cells by EBV may lead to the rapid growth of B cell clones that undergo malignant transformation. The B cell lymphomas are now treated with two forms of immunotherapy—EBV-specific autologous T cell clones, and anti-B cell monoclonal antibody (Rituximab). These are treatments that could be quantitatively explored in the murine immunosuppressive model of radiation and latent virus infection. Use of this model would permit the animal trials of additional modes of immunotherapy, such as the use of designer fusion proteins containing tumor-specific antibody and B-cell toxic reagents. Thus, this space research on irradiated and virus-infected mice may yield valuable information for the treatment of lymphomas in humans.
Task Progress:	We have been able to demonstrate that radiation and latent virus infection in a mouse model produce reactivation of virus infection and altered T-cell cytotoxicity responses. These findings have not been previously demonstrated, and they demonstrate the power of this model system to quantitate potentially harmful effects of space conditions on the ability of the host to resist the development of chronic viral infection with the potential for development of malignancies.
Bibliography Type:	Description: (Last Updated: 04/27/2011)
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Presentation	Shearer, W. T. "“Antarctica Winter Isolation Alters the Pro-Inflammatory/Anti-Inflammatory Cytokine Balance in Humans Experiencing Reactivation of Latent Viruses: Implications for Chronic Viral Infection and Development of Malignancy“ " Jan-2003
Presentation	Shearer, W. T. "“Antarctica Winter Isolation Alters the Pro-Inflammatory: Anti-Inflammatory Cytokine Balance in Humans Experiencing Reactivation of Latent Viruses: Implications for Chronic Viral Infection and Development of Malignancy“ " Jun-2003